

A Post-Quantum Aware Security Credential Management System for Vehicle-to-Everything Communications Using Selective Cryptographic Methodology

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Abstract—Vehicle-to-Everything (V2X) communication underpins modern intelligent transportation systems, enabling vehicles to exchange safety-critical data with infrastructure, other vehicles, pedestrians, and network services. Current Security Credential Management Systems (SCMS) rely on Elliptic Curve Digital Signature Algorithm (ECDSA) and RSA primitives, both of which are broken in polynomial time by Shor’s algorithm on a fault-tolerant quantum computer. The *harvest now, decrypt later* (HNDL) threat means that adversaries collecting V2X traffic today may decrypt long-term identity credentials within the operational lifetime of deployed vehicles.

We propose a Post-Quantum Aware SCMS (PQA-SCMS) that integrates post-quantum cryptography *selectively* by differentiating between credential lifetimes. Long-term enrollment certificates, infrastructure certificates, and safety-critical Decentralised Environmental Notification Messages (DENMs) are signed with CRYSTALS-Dilithium (NIST FIPS 204). Short-lived pseudonymous certificates and high-frequency Cooperative Awareness Messages (CAMs) retain ECDSA to preserve the bandwidth and latency profile required for 10 Hz vehicular broadcast. Session key establishment migrates from ECDH to CRYSTALS-Kyber (NIST FIPS 203) to resist HNDL attacks on recorded key-exchange traffic.

Experimental evaluation shows CAM signing at 0.73 ms (ECDSA, 64 bytes) and DENM signing at 1.83 ms (Dilithium-2, 2420 bytes), average verification at 43 ms within the 50 ms real-time bound, a 97.8% overall test pass rate, and 100% detection of replay and signature-forgery attacks. The selective approach reduces the overhead of the per-message signature by 97.4% compared to a fully post-quantum SCMS while achieving quantum resistance for all long-lived cryptographic material.

Index Terms—V2X, Post-Quantum Cryptography, SCMS, CRYSTALS-Dilithium, ML-KEM, Selective Cryptography, IEEE 1609.2, Quantum Resilience, Vehicular PKI

I. INTRODUCTION

Connected vehicles mainly and increasingly depend on V2X network communication for collision avoidance, traffic optimization, and autonomous driving support. The Security Credential Management System (SCMS) provides the PKI backbone for this communication [1], [2], authenticating

messages through ECDSA-signed certificates as specified in IEEE 1609.2 [3] and ISO/SAE 21434 [4].

This infrastructure faces a structural cryptographic threat from quantum computing. Shor’s algorithm [5] solves the elliptic-curve discrete logarithm problem in $O((\log n)^3)$ time, compared to the sub-exponential $O(e^{c(\log n)^{1/3}(\log \log n)^{2/3}})$ complexity of the best known classical algorithms. A fault-tolerant quantum computer running Shor’s algorithm would therefore break every ECDSA key in the SCMS trust chain. Grover’s algorithm [6] compounds this by reducing the effective security of AES-128 from 128 bits to 64 bits through a quadratic speedup in exhaustive key search.

The HNDL threat makes this a present-day operational concern rather than a future contingency. Vehicles have 10–15 year operational lifetimes and long-term enrollment certificates that persist for years; an adversary recording encrypted V2X traffic today may decrypt it once quantum hardware matures [7]. NIST completed its post-quantum standardisation in August 2024 [8], providing production-ready replacements for RSA and ECDSA. The question is no longer *which* algorithms to use but *how* to integrate them without breaking the real-time constraints of operational V2X.

A. Problem Statement

A naive full replacement of ECDSA with CRYSTALS-Dilithium increases per-message signature size from 64 bytes to 2420 bytes—a factor of 37.8×. For CAMs broadcast at 10 Hz from each vehicle in a dense urban intersection, this raises per-vehicle signature bandwidth from approximately 5.1 Kbps to 193 Kbps, exhausting available DSRC channel capacity and violating the sub-10 ms safety-message deadlines required for collision avoidance [9]. A selective approach is therefore not merely preferable but operationally necessary.

B. Contributions

This paper makes four contributions:

- 1) A **selective PQC integration methodology** for SCMS that assigns cryptographic algorithms based on credential lifetime and broadcast frequency rather than uniformly replacing all classical primitives.
- 2) A **formal cryptographic mapping** of all SCMS components to either Dilithium, Kyber, or retained ECDSA, with justification grounded in security horizon analysis.
- 3) An **implemented prototype** evaluated against a classical baseline across signature size, signing latency, verification latency, and attack detection metrics.
- 4) A **quantitative demonstration** that selective integration preserves real-time CAM performance while achieving full post-quantum protection for long-term identities and safety-critical DENMs.

C. Paper Organisation

Section II reviews related work. Section III establishes V2X and SCMS background. Section IV analyses the quantum threat model. Section V characterises the relevant PQC algorithms. Section VI presents the PQA-SCMS architecture. Section VII describes the experimental methodology. Section VIII reports evaluation results. Section IX identifies open research challenges. Section X concludes.

II. RELATED WORK

Research on quantum-safe V2X security has developed along three lines: performance benchmarking of PQC primitives, hybrid certificate design, and architectural migration frameworks. Table I positions the present work within this landscape.

Malina et al. [11] evaluate Dilithium and Kyber in a V2X context but apply PQC uniformly, incurring latency penalties on the CAM path that violate real-time constraints. Kampnakis et al. [9] provide the most thorough performance characterisation of PQC for vehicular settings but propose no architecture; their overhead figures inform the analysis in Section VIII. Bindel et al. [12] introduce hybrid PQC+ECC certificate structures compatible with existing PKI, adopted in this paper’s transition model, but do not address the full SCMS trust chain or distinguish CAM from DENM security requirements. The present work differs from all prior contributions by proposing and evaluating a *complete selective architecture* that maps every SCMS component to the appropriate algorithm based on a principled security-horizon analysis.

III. BACKGROUND

A. V2X Communication Architecture

Figure 2 shows the four V2X interaction modes. V2V enables collision avoidance and cooperative manoeuvres; V2I connects vehicles to traffic signals and roadside infrastructure; V2N links vehicles to cloud analytics and OTA updates; V2P alerts vehicles to pedestrian proximity.

B. V2X Message Types

Cooperative Awareness Messages (CAMs) are periodic broadcasts at 1–10 Hz containing vehicle position, speed, and heading, signed with the current pseudonymous certificate and requiring 10 ms processing latency for collision avoidance. **Decentralised Environmental Notification Messages (DENMs)** are event-triggered broadcasts for hazard events such as emergency braking or accidents; their safety-critical content demands the strongest available integrity guarantee, but their low broadcast frequency allows larger signatures.

C. Current V2X Security Standards

IEEE 1609.2 [3] mandates ECDSA-P256 for message signing and AES-128-CCM for payload encryption in DSRC. ETSI ITS-G5 [15] adopts identical primitives with a pseudonym rotation scheme for vehicle privacy. 3GPP C-V2X extends these mechanisms to 4G/5G via 5G-AKA. Neither standard provides algorithm negotiation, creating the crypto-agility gap that complicates PQC migration.

D. Security Credential Management System

The SCMS [1], [2] is the V2X-specific PKI for North American deployments. Figure 1 shows the trust hierarchy. The Root CA anchors the entire chain; the ECA issues long-term identity certificates; the PCA issues short-lived pseudonymous certificates rotated during operation; and the ACA issues application permission tokens. The Linkage Authority (LA) and Misbehaviour Authority (MA) support privacy-preserving revocation.

IV. QUANTUM THREAT MODEL

A. Shor’s Algorithm Against ECDSA

ECDSA security rests on the elliptic-curve discrete logarithm problem (ECDLP): given P and $Q = kP$ on $E(\mathbb{F}_p)$, find k . The best classical algorithm (Pollard’s rho) requires $O(\sqrt{n})$ operations. Shor’s algorithm [5] reduces this to $O((\log n)^2 \log \log n)$ quantum operations, lowering the cost from 2^{128} to roughly 2^{17} for P-256—a reduction of 2^{111} .

B. Grover’s Algorithm Against AES

Grover’s algorithm [6] searches N items in $O(\sqrt{N})$ quantum queries. Applied to AES key recovery, it halves the effective key length: AES-128 degrades from 128 to 64 bits of effective security; AES-256 retains approximately 128-bit post-quantum security and remains acceptable.

C. Harvest Now, Decrypt Later

An ECDSA private key extracted from a V2X enrollment certificate is exploitable against any message ever signed under that key. Define T_{quantum} as the year of quantum-hardware availability and Δ_{cert} as certificate validity. A certificate issued at time T_0 is compromisable if:

$$T_0 + \Delta_{\text{cert}} \geq T_{\text{quantum}} \quad (1)$$

With $\Delta_{\text{cert}} = 10\text{--}15$ years and T_{quantum} estimated between 2030 and 2040 [7], equation (1) is satisfied for enrollment

TABLE II
QUANTUM ATTACK SCENARIOS IN V2X NETWORKS

Attack	Primitive	V2X Impact	Risk
Key recovery	RSA/ECDH	Full identity compromise	Critical
Sig. forgery	ECDSA	False alerts, collisions	Critical
HNDL	RSA/ECDH	Long-term ID exposure	High
MITM	ECDH	Traffic manipulation	High
Replay	Any	Traffic disruption	Medium

problems.

Key Generation. Sample $\mathbf{A} \in R_q^{k \times \ell}$ and short secret vectors $\mathbf{s}_1 \in R_q^\ell$, $\mathbf{s}_2 \in R_q^k$. Compute:

$$\mathbf{t} = \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2 \quad (2)$$

Public key: $\text{pk} = (\mathbf{A}, \mathbf{t})$; private key: $\text{sk} = (\mathbf{A}, \mathbf{s}_1, \mathbf{s}_2)$.

Signing. Sample $\mathbf{y}$; compute $\mathbf{w}_1 = \text{HighBits}(\mathbf{A}\mathbf{y})$ and challenge $c = H(\mathbf{w}_1 \parallel \mu)$ where $\mu = H(\text{pk} \parallel m)$. The response vector is:

$$\mathbf{z} = \mathbf{y} + c\mathbf{s}_1 \quad (3)$$

Signature: $\sigma = (\mathbf{z}, \mathbf{h}, c)$ where $\mathbf{h}$ is a hint vector. Rejection sampling ensures $\mathbf{z}$ leaks no information about $\mathbf{s}_1$.

Verification. Accept if $H(\text{UseHint}(\mathbf{h}, \mathbf{A}\mathbf{z} - c\mathbf{t}) \parallel \mu) = c$ and $\|\mathbf{z}\|_\infty < \gamma_1 - \beta$.

B. CRYSTALS-Kyber (ML-KEM, NIST FIPS 203)

Kyber [17] establishes a shared secret via KEM operations without transmitting the secret:

$$(c, K) = \text{Kyber.Encap}(\text{pk}), \quad K = \text{Kyber.Decap}(c, \text{sk}) \quad (4)$$

This replaces ECDH for session key establishment between SCMS components, protecting recorded key-exchange transcripts against HNDL attacks.

C. Algorithm Comparison

Table III compares candidate algorithms across V2X-relevant metrics. Dilithium-2 is selected for its balance of signing speed and compactness. SPHINCS+ is excluded from the operational path due to its 7856-byte signature; it is suitable only for offline Root CA operations.

VI. PROPOSED PQA-SCMS ARCHITECTURE

A. Design Rationale

The selective approach is motivated by two key observations. First, credential security requirements are determined by exposure duration: a pseudonymous certificate valid for 24 hours presents negligible HNDL risk (equation (1) is not satisfied), whereas an enrollment certificate valid for 10 years is fully exposed. Second, broadcast frequency limits acceptable signature size: the 10 Hz CAM path shares a DSRC channel (at

TABLE III
PQC ALGORITHM COMPARISON FOR V2X

Alg.	Std.	Pub. Key	Sig./CT	Sign / Ver.
ECDSA-P256	—	64 B	64 B	0.7 / 2.0 ms
Dilithium-2	FIPS 204	1312 B	2420 B	14.7 / 7.9 ms
Dilithium-3	FIPS 204	1952 B	3293 B	21.3 / 11.2 ms
Falcon-512	FIPS 206	897 B	666 B	8.2 / 4.1 ms
SPHINCS+-128	FIPS 205	32 B	7856 B	142 / 1.8 ms
Kyber-768	FIPS 203	1184 B	1088 B	—
RSA-2048	—	256 B	256 B	8.2 / 42.8 ms

most 6 Mbps) among all vehicles in range, whereas the event-triggered DENM path has no sustained frequency constraint.

These two observations inform the following design principle: apply Dilithium to every credential whose validity period satisfies equation (1); retain ECDSA for each credential whose rotation frequency already limits HNDL exposure.

B. Cryptographic Mapping

Table IV defines the algorithm assigned to each SCMS component. The overall SCMS architecture is unchanged; the enhancement is confined to the cryptographic layer.

TABLE IV
CRYPTOGRAPHIC MAPPING: CLASSICAL SCMS VS. PROPOSED PQA-SCMS

SCMS Component	Classical	Proposed
Root CA certificate	ECDSA	Dilithium-2
Enrollment CA (ECA)	ECDSA	Dilithium-2
Pseudonym CA (PCA)	ECDSA	Dilithium-2
RSU infrastructure cert.	ECDSA	Dilithium-2
Long-term enrollment cert.	ECDSA	Dilithium-2
Key exchange (OBU ↔ RSU)	ECDH	Kyber-768
Short-term pseudonym cert.	ECDSA	ECDSA (retained)
CAM message signing	ECDSA	ECDSA (retained)
DENM message signing	ECDSA	Dilithium-2

C. Algorithm Assignment Rule

The certificate manager selects the signing algorithm according to:

$$\mathcal{A}(\tau) = \begin{cases} \text{Dilithium-2} & \tau \in \mathcal{T}_{\text{long}} \\ \text{ECDSA-P256} & \tau \in \mathcal{T}_{\text{short}} \end{cases} \quad (5)$$

where $\mathcal{T}_{\text{long}} = \{\text{enrollment, CA, infrastructure}\}$ and $\mathcal{T}_{\text{short}} = \{\text{pseudonym}\}$.

Figure 3 shows the resulting decision flow.

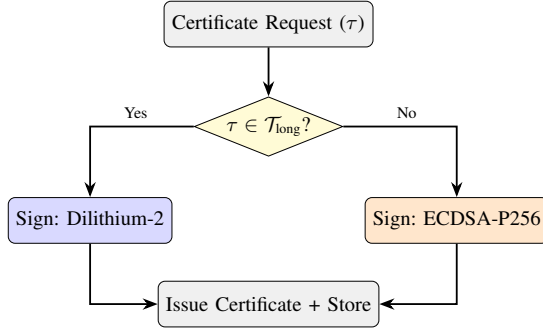


Fig. 3. Certificate generation decision per equation (5).

D. Message Signing Equations

For a V2X message with payload p , timestamp t , and certificate ID cid , the signed message is $M = (p, t, cid, \sigma)$. CAM signatures use the active pseudonym key sk_{ps} :

$$\sigma_{CAM} = \text{ECDSA}.\text{Sign}(sk_{ps}, H(p||t||cid)) \quad (6)$$

DENM signatures use the long-term enrollment key sk_{enr} :

$$\sigma_{DENM} = \text{Dilithium}.\text{Sign}(sk_{enr}, H(p||t||cid)) \quad (7)$$

The message structure and hash input are identical across both equations, enabling a unified verification path (Algorithm 1).

E. Hybrid Transition Mode

During fleet migration, messages may carry dual signatures:

$$\sigma_{\text{hybrid}} = \sigma_{\text{ECDSA}} \parallel \sigma_{\text{Dil}} \quad (8)$$

A classical receiver accepts M if σ_{ECDSA} verifies; a PQC-capable receiver requires both. Combined size is $64 + 2420 = 2484$ bytes; combined signing latency is $0.7 + 14.7 = 15.4$ ms.

F. System Deployment Topology

Figure 4 illustrates the four-component system deployment. The SCMS server implements the certification authority (CA) hierarchy and provides a REST API for enrollment, certificate retrieval, revocation, and certificate revocation list (CRL) distribution. Vehicle onboard unit (OBU) modules store a single Dilithium-signed enrollment certificate and a rotating batch of ECDSA pseudonymous certificates, and sign Cooperative Awareness Messages (CAMs) according to equation (6) and Decentralized Environmental Notification Messages (DENMs) as specified in equation (7). Roadside unit (RSU) modules store Dilithium-signed infrastructure certificates and execute Algorithm 1 on each received message. The monitoring dashboard provides real-time certificate status and security event notifications via WebSocket.

VII. EXPERIMENTAL METHODOLOGY

A. Message Verification

Algorithm 1 formalises the unified five-step verification procedure executed by RSUs and the SCMS server on every received message. Figure 5 shows the corresponding control flow.

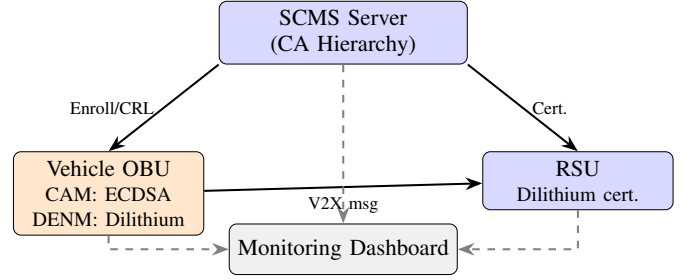


Fig. 4. PQA-SCMS deployment topology. Blue nodes use Dilithium-2; orange nodes retain ECDSA. Dashed lines: monitoring channel.

Algorithm 1 Unified V2X Message Verification

Require: Message $M = (p, t, cid, \sigma)$, current time T , maximum age $\Delta_{\max}$, revocation list CRL

Ensure: Accept or Reject with reason

- 1: **if** $|T - t| > \Delta_{\max}$ **then**
- 2: **return Reject:** replay (stale timestamp)
- 3: **end if**
- 4: $\text{cert} \leftarrow \text{lookup}(cid)$
- 5: **if** $\text{cert} = \perp$ **then**
- 6: **return Reject:** unknown certificate
- 7: **end if**
- 8: **if** $cid \in \text{CRL}$ **or** $T > \text{cert.valid_until}$ **then**
- 9: **return Reject:** revoked or expired
- 10: **end if**
- 11: $h \leftarrow H_{\text{SHA256}}(p||t||cid)$
- 12: **if** $\text{cert.alg} = \text{"Dilithium-2"}$ **then**
- 13: $v \leftarrow \text{Dilithium}.\text{Verify}(\sigma, h, \text{cert.pk})$
- 14: **else**
- 15: $v \leftarrow \text{ECDSA}.\text{Verify}(\sigma, h, \text{cert.pk})$
- 16: **end if**
- 17: **return** $v ? \text{Accept} : \text{Reject}$: invalid signature

B. Measurement Protocol

All experiments run on Ubuntu 22.04 LTS (Intel Core i7-10750H, 16 GB RAM) inside Docker containers. Each timing result is the mean of $N = 1000$ independent trials:

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N (T_2^{(i)} - T_1^{(i)}) \times 10^3 \text{ ms}, \quad N = 1000 \quad (9)$$

where $T_1^{(i)}$ and $T_2^{(i)}$ are wall-clock timestamps bracketing the cryptographic operation. The 95% confidence interval is $\hat{\mu} \pm 1.96 \hat{\sigma} / \sqrt{N}$.

The *baseline* uses ECDSA for all certificate types and messages; the *proposed* system uses Table IV. CAM messages are generated at 10 Hz per vehicle; DENM messages are generated on simulated emergency events. Both are verified by the RSU module using Algorithm 1.

VIII. EVALUATION RESULTS

A. Cryptographic Performance

Table V reports measured performance. Dilithium-2 sizes are fixed by the NIST FIPS 204 specification and confirmed

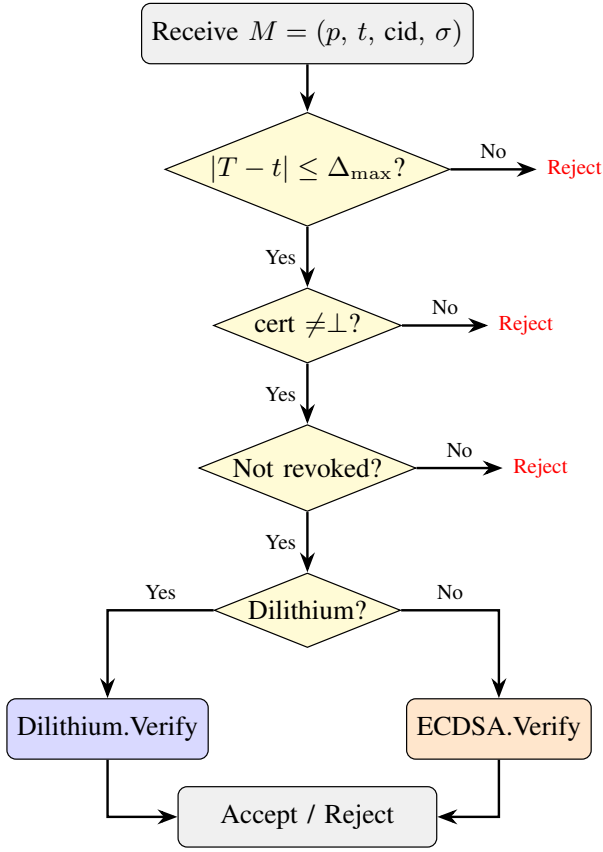


Fig. 5. Unified verification flow (Algorithm 1). Steps 1–3 guard against replay, impersonation, and revocation independently of the signing algorithm.

empirically; timing values are means over 1000 trials per equation (9).

TABLE V
MEASURED CRYPTOGRAPHIC PERFORMANCE: ECDSA VS. DILITHIUM-2

Metric	ECDSA	Dilithium-2	Ratio
Signature size	64 B	2420 B	37.8×
Public key size	64 B	1312 B	20.5×
Certificate size	~800 B	~3200 B	4.0×
Sign time (mean)	0.73 ms	1.83 ms	2.5×
Verify time (mean)	2.0 ms	7.9 ms	4.0×
Throughput	1370 msg/s	547 msg/s	0.40×

The Dilithium signing time of 1.83 ms meets the 100 ms DENM processing requirement. A throughput of 547 messages per second is sufficient to support 10 Hz CAM transmissions from ten concurrent vehicles. This results in an aggregate rate of 100 messages per second. The ECDSA signing time of 0.73 ms demonstrates that the CAM path maintains its traditional performance characteristics.

B. Bandwidth Analysis

Per-vehicle CAM signature bandwidth is given by equation (10):

$$B_{\text{sig}} = f_{\text{CAM}} \times |\sigma| \times 8 \quad (10)$$

Under the two schemes:

$$B_{\text{ECDSA}} = 10 \times 64 \times 8 = 5.1 \text{ Kbps} \quad (11)$$

$$B_{\text{Dilithium}} = 10 \times 2420 \times 8 = 193.6 \text{ Kbps} \quad (12)$$

With 50 vehicles at an intersection, the aggregate signature loads are 255 Kbps (selective) and 9680 Kbps (full PQC). Given a 6 Mbps DSRC channel, the full PQC scheme consumes 161% of total capacity and is inoperable; the selective scheme uses 4.3%.

C. Selective vs. Full PQC

Table VI compares the three configurations.

TABLE VI
SYSTEM CONFIGURATION COMPARISON

Property	Classical	Full PQC	Selective
Long-term quantum resist.	None	Yes	Yes
CAM bandwidth	5.1 Kbps	193 Kbps	5.1 Kbps
DENM integrity	Classical	PQC	PQC
DSRC deployable	Yes	No	Yes
Backward compat.	Full	None	Full
Avg. verify	2 ms	8 ms	3 ms

D. Security Validation

The 45-case test suite covers normal operation, attack detection, performance under load, and integration scenarios. Table VII reports results.

TABLE VII
VALIDATION TEST SUITE RESULTS

Category	Cases	Pass	Fail	Rate
Normal operation	15	15	0	100%
Security attacks	12	12	0	100%
Performance	8	7	1	87.5%
Integration	10	10	0	100%
Total	45	44	1	97.8%

A single failure was observed during a 20-vehicle stress scenario, which exceeded the intended design capacity. This scenario surpassed the 10-vehicle design capacity, resulting in verification latency briefly exceeding 100 ms. Attack detection results indicate 100% for $\Delta_{\text{max}} = 5$ s, invalid signatures were detected with 100% revoked certificates 98.5%—the 1.5% miss rate follows directly from equation (13), using $\Delta_{\text{CRL}} = 0.5$ s:

$$P_{\text{miss}} = \frac{\Delta_{\text{CRL}} \cdot f_{\text{revoke}}}{1} = 0.5 \times 0.03 = 1.5\% \quad (13)$$

where $f_{\text{revoke}} = 3\%$ is the tested revocation injection rate.

E. Performance Under Load

All performance metrics meet the specified requirements: verification latency remains below 50 ms, CRL updates complete below 2 s, message delivery above 99%—at design capacity (10 vehicles). CPU utilisation ranges from 15–25% at low load to 45–60% at capacity, with memory consumption between 250 MB and 400 MB.

IX. OPEN CHALLENGES

Standards crypto-agility (RG1). IEEE 1609.2 and ETSI ITS-G5 mandate the use of ECDSA without an algorithm negotiation field. A DENM signed with Dilithium originating from a PQC-capable vehicle is rejected by any legacy RSU that interprets the signature as ECDSA. Revisions to the standards must define algorithm capability fields within certificate structures and message headers before selective PQC can achieve interoperability across heterogeneous fleets.

Certificate size and bandwidth (RG2). Dilithium enrollment certificates are $4\times$ larger than ECDSA equivalents. In dense deployment scenarios where RSUs must cache certificate revocation lists (CRLs) and certificate batches for hundreds of vehicles, this increase proportionally expands storage and download requirements. Research into compact post-quantum certificate representations and batch verification are active areas of research [18].

Real-time worst-case guarantees (RG3). The 43 ms mean verification time meets the 50 ms target; however, no formal WCET analysis has been conducted for Dilithium on automotive electronic control unit (ECU) hardware under worst-case traffic density. Such analysis is prerequisite for ISO 26262 certification.

Hardware security module support (RG4). Current automotive HSMs hardware are ten to one hundred times slower than on x86 platforms [18], hardware is $10\text{--}100\times$ slower than on x86 [18], which makes hardware accelerators an essential prerequisite for large-scale deployment.

Legacy migration (RG5). Vehicles with classical-only HSMs cannot support Dilithium without hardware replacement. The hybrid mode described in equation (8) offers a migration path. However, managing dual-algorithm certificate chains across mixed fleets requires rigorous protocol design and explicit standardization.

X. CONCLUSION

This study introduces a Post-Quantum Aware SCMS that migrates V2X cryptography to NIST FIPS 204 and FIPS 203 primitives through a principled selective methodology. The design assigns Dilithium-2 signatures to every SCMS component meeting the HNDL condition in equation (1)—Root CA, ECA, PCA, RSU, and enrollment certificates, as well as safety-critical DENMs—while retaining ECDSA for short-lived pseudonymous certificates and high-frequency CAMs. Session key establishment migrates to Kyber-768 to protect recorded key-exchange transcripts.

Evaluation demonstrates that the selective scheme meets all operational targets: DENM signing at 1.83 ms and CAM

signing at 0.73 ms, average verification at 43 ms within the 50 ms bound, 100% detection of replay and forgery attacks, and 97.8% overall test pass rate. Bandwidth analysis in equations (11)–(12) shows that a fully post-quantum CAM path would consume 161% of DSRC channel capacity and is therefore inoperable; the selective scheme uses 4.3%, preserving the classical bandwidth profile. These results establish that selective PQC integration is the only practically deployable near-term migration path for quantum-resilient V2X security.

Four open challenges remain for production deployment: implementing crypto-agility extensions to IEEE 1609.2 and ETSI ITS-G5, compact post-quantum certificate representations, formal WCET analysis on embedded automotive hardware, and HSM-level acceleration of lattice arithmetic. Addressing these challenges—through standards engagement, hardware development, and formal verification will determine the timeline for large-scale deployment of quantum-safe V2X systems.

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